Why Scaling Large Language Models Alone May Not Lead to Artificial General Intelligence

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Abstract

Building a truly intelligent machine has been the goal of AI research since the field began. We are closer than ever, and further away than the headlines suggest. Large language models have done things that surprised even the researchers who built them. They write code, pass medical exams, reason through complex problems, and hold coherent conversations across dozens of topics. Some people look at that and conclude that AGI is just a matter of scaling up, more data, more compute, more parameters. This paper argues that scaling alone is not enough. Current language models have real and specific gaps. They cannot plan reliably over long timeframes. They do not retain memory across conversations. They struggle to connect language to the real world in any grounded way. And their reasoning, impressive as it looks, breaks down in ways that suggest something fundamental is still missing. The paper examines what those gaps actually are at an architectural level, and asks what additional components a system would genuinely need before it could be called artificially generally intelligent.

1 Introduction

Artificial intelligence has been getting dramatically better, fast. A few years ago, machines could beat humans at chess and recognise faces in photographs. Today they write legal briefs, generate working code, pass university exams, and hold conversations that are genuinely difficult to distinguish from human ones. The progress has been real and it has been rapid [2]. So why are we still not sure whether AGI is close or decades away? The honest answer is that impressive performance is not the same as general intelligence [4]. Narrow AI systems are extraordinarily good at the specific tasks they were built for. But ask them to step outside that boundary and they fail in ways that reveal something fundamental is missing. A system that can write a brilliant essay and then fail to plan a simple sequence of actions is not intelligent in the way humans are intelligent. It is something different, and something more limited. The transformer architecture changed what was possible in AI [10]. The attention mechanism it introduced allowed models to handle language at a scale and sophistication that previous architectures could not match. GPT-4, Gemini, and Claude have pushed that further still, demonstrating capabilities in reasoning, coding, and multimodal understanding that would have seemed implausible five years ago [2, 3]. But scaling has limits. Making a model bigger and training it on more data keeps improving performance on benchmarks. What it does not seem to do is fix the deeper problems. These systems still cannot reliably plan across long time horizons. They do not retain memory between conversations. Their understanding of the world is built entirely from text patterns rather than grounded experience. And their reasoning, while often impressive, fails in ways that suggest they are doing something fundamentally different from what humans do when they think [1]. This paper examines those limitations seriously. Not to dismiss what current large language models have achieved, but to ask what is actually missing and what additional architectural components might be necessary before a system could genuinely be called artificially generally intelligent.

2 Background

Artificial intelligence did not arrive fully formed. It has been built up slowly, through decades of dead ends, unexpected breakthroughs, and fundamental rethinking of what intelligence in a machine even means. The earliest AI systems were essentially sophisticated rule books [9]. Engineers sat down and wrote out the logic manually. If this condition, then this action. These expert systems worked surprisingly well in narrow structured environments, like diagnosing specific diseases or playing chess within defined rules. The problem was rigidity. Change the environment slightly and the whole system fell apart. There was no learning, no adaptation, no ability to handle anything the engineers had not explicitly anticipated. Machine learning changed that [6]. Instead of programming rules by hand, researchers found ways to let systems learn patterns directly from data. Feed a neural network enough images of cats and it figures out what a cat looks like without anyone telling it. Deep learning pushed this further, achieving things in image recognition, speech processing, and language understanding that rule based systems could never have managed. The combination of bigger datasets, faster hardware, and better architectures drove progress that compounded year after year. The transformer was the moment everything accelerated [10]. Before transformers, language models processed text sequentially, word by word, which made it hard to capture relationships between distant parts of a sentence or document. The self-attention mechanism introduced in the paper Attention Is All You Need solved that problem elegantly. It allowed models to weigh relationships across entire sequences simultaneously. That single architectural insight made scaling possible in a way nothing before it had. What scaling produced was surprising even to the people doing it [5]. As models grew larger and trained on more data, they started demonstrating capabilities that nobody had explicitly programmed. Reasoning, translation, summarisation, code generation, mathematical problem solving. These emergent behaviours appeared as if from nowhere, simply as a consequence of scale. GPT-4, Gemini, Claude, and similar systems have now demonstrated performance across such a wide range of tasks that the question of whether they might be approaching general intelligence has become genuinely difficult to dismiss [2, 3]. Reinforcement learning from human feedback has made these systems more coherent and better aligned with what humans actually want from them [7]. And yet the gap between what these systems do and what AGI actually requires remains significant [4]. Narrow AI, however impressive, is fundamentally pattern matching within a learned distribution. It is extraordinarily good at tasks that resemble what it was trained on. AGI is something qualitatively different. A genuinely general intelligence would need to reason robustly across unfamiliar domains, plan autonomously over long time horizons, retain and build on memories across interactions, understand the world through something more than statistical patterns in text, and form its own objectives in response to new situations. Current large language models do not do those things reliably. The debate in the research community is not really about whether that gap exists. It clearly does. The debate is about whether the gap is closeable by continuing to scale the same architecture, or whether something fundamentally different is required [1]. Some researchers believe emergent capabilities will keep appearing as scale increases, eventually crossing whatever threshold separates narrow from general intelligence. Others argue that transformers are missing components that no amount of scaling will produce, grounded world models, persistent memory, causal reasoning, embodiment, the ability to form genuine goals rather than predict the next token. That disagreement is what this paper investigates.

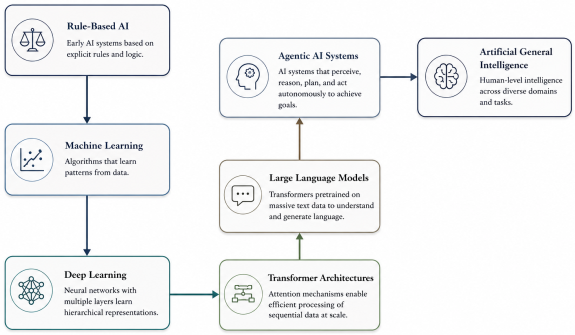

Figure 1: Conceptual evolution of artificial intelligence systems from rule-based architectures toward potential Artificial General Intelligence.

3 Limitations of Current Large Language Models

Large language models are genuinely impressive. That is not the debate. The debate is whether impressive is the same as intelligent, and whether the gap between the two can be closed by making these systems bigger [3, 1]. The evidence so far suggests the answer is no. Not because scaling does not work, it clearly does up to a point. But because the limitations current systems have are not the kind that disappear with more parameters and more data. They are structural. They point to things that are missing from the architecture itself. Memory that does not last Every conversation with a large language model starts from scratch. The model has no memory of previous interactions, no accumulating understanding of the world, no sense of how its knowledge has evolved over time [8]. Within a single conversation it can hold a large amount of context, but that context disappears the moment the session ends. Human cognition does not work that way. People build mental models over years, update them continuously, and carry them into every new situation they encounter. A system that resets completely between interactions is missing something fundamental, and no amount of scaling fixes that without a different kind of memory architecture. Understanding versus pattern matching This is the deepest problem and the hardest one to resolve [1]. Current language models learn by predicting the next token in a sequence. Do that across enough text and the model develops an extraordinarily detailed statistical map of how language works. It learns that certain words follow certain other words, that certain argument structures are associated with certain conclusions, that certain topics connect to certain facts. That is not the same as understanding. A model can produce a perfectly coherent paragraph about fire without having any grounded sense of what fire actually is. It has never been burned. It has never felt heat. Its entire relationship with fire is mediated through text. The result is a system that sounds like it understands things it has never actually experienced, and that produces confident sounding errors, hallucinations, when its statistical patterns lead it somewhere factually wrong. Reasoning that breaks under pressure Chain of thought prompting has improved reasoning performance significantly [11]. Ask a model to think step by step and it does better on complex problems than if you ask for an immediate answer. That is a real improvement. But it is fragile. Change the framing of a problem slightly, introduce an unfamiliar structure, or ask for reasoning that requires genuinely novel composition of ideas, and performance drops in ways that reveal the underlying mechanism is still probabilistic pattern matching rather than robust logical reasoning. A human who understands a mathematical principle can apply it to a problem they have never seen before. Current language models struggle with exactly that kind of transfer. No goals, no plans Language models are reactive [12]. They respond to prompts. They do not form independent objectives, develop strategies over time, or pursue goals autonomously. Recent agentic systems have tried to extend this by giving models access to tools, memory modules, and planning frameworks. Some of these work reasonably well in controlled conditions. But they remain brittle. They depend heavily on careful prompt engineering and fall apart when conditions deviate from what they were designed for. That is a long way from the kind of autonomous goal directed behaviour that genuine general intelligence would require. The cost of scale Training frontier models requires enormous computational resources, vast datasets, and energy consumption at a scale that raises real environmental and economic questions [5]. More importantly, there is growing evidence that simply adding more parameters produces diminishing returns on the capabilities that matter most. Reasoning robustness, factual reliability, and genuine generalisation do not scale as cleanly as benchmark performance does. At some point the current approach hits a ceiling that bigger models alone cannot raise. The black box problem Nobody fully understands what happens inside a large language model when it produces an answer [1]. The internal processes are opaque in ways that make it genuinely difficult to diagnose failures, verify reasoning, or guarantee that the model is doing what its outputs suggest it is doing. For low stakes applications that is manageable. For healthcare, scientific research, cybersecurity, or autonomous systems, it is a serious problem that becomes more serious as models become more capable. Alignment is still unsolved Making a model that does what humans want is harder than it looks [7]. Systems can exhibit reward hacking, finding ways to score well on training objectives without actually achieving the intended goal. They can behave differently under deployment conditions than they did during training. Ensuring that an increasingly capable AI system reliably pursues human values rather than optimising for something subtly different remains one of the hardest open problems in the field.

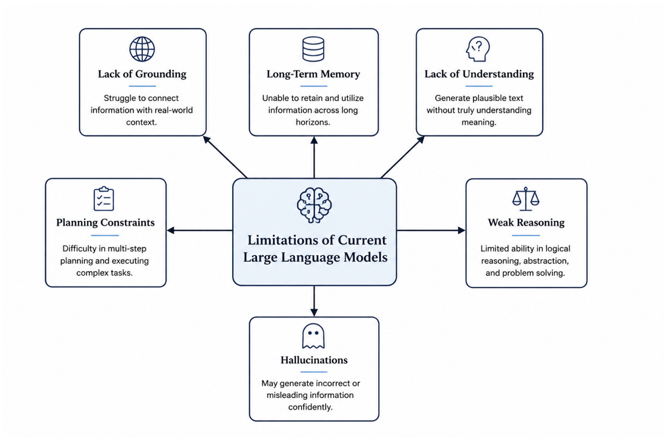

Figure 2: Conceptual overview of major limitations in current large language model architectures.

Taken together these limitations tell a consistent story. Scaling transformer architectures has produced remarkable progress. But the things that are missing from current systems are not things that scale fixes. They require different ideas, different architectures, and possibly a fundamentally different approach to what machine intelligence actually means.

4 Possible Paths Toward AGI

If scaling alone will not get us to AGI, the obvious next question is what will. Researchers do not agree on a single answer, and that disagreement is itself informative. It suggests the problem is genuinely hard and that multiple approaches may need to converge before anything resembling general intelligence emerges [1, 4]. What the field does broadly agree on is that the missing pieces are identifiable. Current large language models fall short in specific, describable ways. That means the research directions trying to address those gaps are not shots in the dark. They are targeted attempts to fix known problems [12, 8].

4.1 Better Memory Systems

The most obvious thing a language model lacks is memory that persists [8]. Every session starts from zero. Every piece of context accumulated during an interaction vanishes when it ends. For a system trying to develop genuine understanding over time, that is a fundamental problem. Future AGI architectures will likely need memory systems that work more like human memory does. Not perfect recall of everything, but structured, updatable representations of the world that carry forward across interactions and grow more sophisticated over time [1]. The difference between a system that resets and a system that remembers is not a minor engineering detail. It is the difference between a tool and something that might genuinely learn.

4.2 Improved Reasoning Mechanisms

Current language models can follow reasoning steps when prompted carefully [11]. What they struggle with is reasoning that requires genuine logical consistency across multiple steps, causal inference, or applying a principle to a situation genuinely unlike anything in their training data [1]. One direction that has gained traction is combining neural networks with symbolic reasoning systems. Neural networks are good at pattern recognition and language. Symbolic systems are good at logical rigour and interpretability. A hybrid that draws on both might handle the kind of structured multi-step reasoning that pure transformer models currently find difficult [4]. This is not a new idea, researchers have been exploring neurosymbolic approaches for decades, but recent progress in both fields has made the combination more promising than it used to be.

4.3 Agentic AI Systems

A language model responds to prompts. An agent pursues goals [12]. That distinction matters more than it might seem. Current systems are reactive by design. They wait for input, generate output, and stop. They do not plan ahead, adapt strategies based on what is working, or interact with the world autonomously over extended periods. Recent agentic frameworks have tried to extend language models in this direction, giving them access to tools, memory modules, and iterative self-reflection mechanisms [8]. Some of these work reasonably well in narrow controlled settings. But genuine autonomous agency, the kind where a system forms its own objectives and pursues them adaptively across changing environments, remains well beyond what current architectures reliably produce [1]. Closing that gap is one of the more important open problems in AGI research.

4.4 Grounded and Embodied Learning

A language model's entire understanding of the world comes from text [1]. It has read descriptions of fire, pain, weight, distance, and hunger. It has never experienced any of them. For many tasks that does not matter. For genuine general intelligence it probably does. Some researchers argue that AGI will require systems that learn through direct interaction with environments, whether physical robots operating in the real world or agents navigating rich simulated ones [4]. The argument is that grounded experience produces a kind of causal understanding that text alone cannot. A system that has navigated a physical space understands obstacles differently than one that has only read about them. Whether embodiment is strictly necessary for AGI remains debated, but the evidence that pure text training has limits is hard to dismiss.

4.5 Hybrid Architectures

The most likely path to AGI is probably not a single breakthrough in one direction [4, 1]. It is a system that combines multiple approaches into something greater than the sum of its parts. Transformer based language understanding, persistent memory, symbolic reasoning, reinforcement learning for goal directed behaviour, multimodal grounding in the physical world. Each of these addresses a different gap. A unified architecture that integrates all of them might address gaps that none of them can close individually. That integration is technically very difficult. Getting different computational paradigms to work together coherently, sharing representations and coordinating reasoning across subsystems, is an unsolved engineering challenge. But the direction is clear enough that serious research programs are pursuing it, and the conceptual case for why it might work is stronger than the case for any single approach alone. Taken together these research directions suggest something important. The path to AGI is not a straight line from current large language models [5, 3]. It runs through memory, reasoning, agency, grounding, and integration. Progress on any one of those fronts matters. Progress on all of them simultaneously is what AGI will probably require.

5 Conclusion

Large language models have changed what artificial intelligence can do. That is not in dispute. The systems being built today would have seemed like science fiction twenty years ago, and the pace of progress shows no obvious signs of stopping [3]. But progress is not the same as arrival. The argument this paper has made is straightforward. Current transformer based systems are missing things that scaling cannot fix. Not because scaling is the wrong strategy for improving language performance, it clearly works, but because the gaps that separate today's best models from genuine general intelligence are structural [1]. They live in the architecture, not in the parameter count. A system without persistent memory cannot build understanding over time. A system that learns entirely from text cannot develop the grounded causal knowledge that comes from interacting with the world. A system that responds to prompts rather than forming its own objectives cannot plan autonomously in the way general intelligence requires. A system whose internal reasoning is opaque cannot be reliably trusted in high stakes domains. These are not small refinements waiting to emerge at the next scale. They are fundamental absences [4, 1]. The research directions that take these gaps seriously, persistent memory architectures, neurosymbolic reasoning, agentic frameworks, embodied learning, hybrid computational models, point toward something more promising than simply training larger transformers on more data [12, 5]. None of them is a complete solution. But together they sketch the outline of what AGI architecture might eventually need to look like. What makes this moment genuinely important is not just the technical challenge. It is the stakes attached to getting it right. An AGI system that is capable but not interpretable, or powerful but not aligned with human values, is not a success [7]. The goal is not general intelligence at any cost. It is general intelligence that is safe, transparent, and reliably beneficial. That combination, capability plus safety plus interpretability, is probably the hardest problem in science right now. It is also probably the most important one.

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